

Tissue	GSE	Samples	Genes	Classes	Shannon entropy
Lung	GSE40419	164 [87 tumor, 77 normal]	58735	2	0.997

GSE40419 - With the 10 Top Genes & Hyperparameter Optimization						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.906±0.008	0.910±0.007	11.335±1.970	0.115±0.010	0.895±0.009	0.919±0.015
SVM	0.974±0.005	0.975±0.005	37.142±16.016	0.025±0.005	0.976±0.004	0.970±0.009
NB	0.976±0.000	0.977±0.000	25.359±0.000	0.011±0.000	0.989±0.000	0.961±0.000
KNN	0.976±0.000	0.977±0.000	74.308±0.000	0.034±0.000	0.966±0.000	0.987±0.000
LR	0.982±0.000	0.983±0.000	38.038±0.000	0.011±0.000	0.989±0.000	0.974±0.000
MLP	0.965±0.010	0.967±0.008	29.099±16.557	0.031±0.013	0.970±0.012	0.959±0.018
RF	0.966±0.007	0.969±0.006	19.952±4.877	0.019±0.007	0.982±0.007	0.948±0.010
DUM	0.530±0.000	0.693±0.000	1.000±0.000	0.000±0.000	1.000±0.000	0.000±0.000

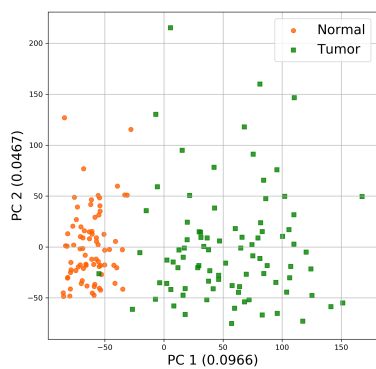
GSE40419 - With all Genes & Hyperparameter Optimization						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.942±0.009	0.945±0.009	26.934±16.439	0.073±0.010	0.930±0.009	0.957±0.016
SVM	0.988±0.000	0.988±0.000	0.000±0.000	0.023±0.000	0.977±0.000	1.000±0.000
NB	0.634±0.000	0.744±0.000	1.284±0.000	0.000±0.000	1.000±0.000	0.221±0.000
KNN	0.939±0.000	0.939±0.000	0.000±0.000	0.115±0.000	0.885±0.000	1.000±0.000
LR	0.988±0.000	0.988±0.000	0.000±0.000	0.023±0.000	0.977±0.000	1.000±0.000
MLP	0.711±0.024	0.760±0.017	1.879±0.232	0.247±0.055	0.868±0.035	0.532±0.063
RF	0.972±0.008	0.973±0.007	39.086±16.171	0.031±0.009	0.969±0.009	0.974±0.013
DUM	0.530±0.000	0.693±0.000	1.000±0.000	0.000±0.000	1.000±0.000	0.000±0.000

GSE40419 - With all Genes & without Hyperparameter Optimization						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.943±0.009	0.945±0.008	26.230±13.990	0.073±0.014	0.930±0.013	0.958±0.016
SVM	0.970±0.000	0.970±0.000	0.000±0.000	0.057±0.000	0.943±0.000	1.000±0.000
NB	0.634±0.000	0.744±0.000	1.284±0.000	0.000±0.000	1.000±0.000	0.221±0.000
KNN	0.939±0.000	0.939±0.000	0.000±0.000	0.115±0.000	0.885±0.000	1.000±0.000
LR	0.988±0.000	0.988±0.000	0.000±0.000	0.023±0.000	0.977±0.000	1.000±0.000
MLP	0.518±0.040	0.572±0.041	1.051±0.138	0.956±0.193	0.609±0.056	0.416±0.053
RF	0.978±0.006	0.979±0.006	60.730±19.053	0.029±0.009	0.971±0.009	0.985±0.010
DUM	0.530±0.000	0.693±0.000	1.000±0.000	0.000±0.000	1.000±0.000	0.000±0.000

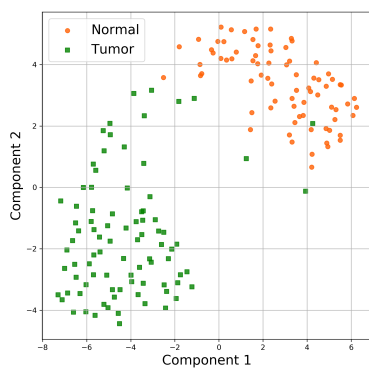
GSE40419 - With the 10 Top Genes & Hyperparameter Optimization & Class weight						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.907±0.009	0.910±0.009	12.112±2.052	0.118±0.014	0.891±0.013	0.925±0.012
SVM	0.973±0.006	0.974±0.006	36.311±16.325	0.026±0.006	0.975±0.005	0.969±0.009
LR	0.982±0.000	0.983±0.000	38.038±0.000	0.011±0.000	0.989±0.000	0.974±0.000
RF	0.965±0.008	0.968±0.007	22.735±5.212	0.027±0.011	0.975±0.010	0.955±0.009

GSE40419 - With the 10 Top Genes & Hyperparameter Optimization & SMOTE						
Classifier	Accuracy	F1-score	Lr+	Lr-	Sensitivity	Specificity
DT	0.910±0.007	0.908±0.007	13.481±2.587	0.120±0.010	0.888±0.009	0.932±0.012
SVM	0.973±0.006	0.973±0.006	38.295±15.211	0.027±0.008	0.974±0.008	0.972±0.007
NB	0.971±0.000	0.972±0.000	21.500±0.000	0.012±0.000	0.989±0.000	0.954±0.000
KNN	0.977±0.000	0.977±0.000	87.818±0.000	0.034±0.000	0.966±0.000	0.989±0.000
LR	0.983±0.000	0.983±0.000	43.000±0.000	0.011±0.000	0.989±0.000	0.977±0.000
MLP	0.964±0.011	0.964±0.011	33.243±20.000	0.037±0.017	0.964±0.017	0.964±0.014
RF	0.970±0.006	0.970±0.006	25.841±7.376	0.020±0.007	0.981±0.007	0.959±0.011
DUM	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000	0.000±0.000

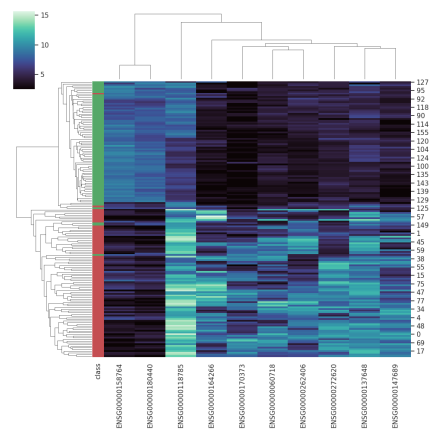
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(a) PCA GSE40419



(b) t-SNE GSE40419



(c) HeatMap GSE40419 - Top 10 Genes

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